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mtu NautIQ

BlueVision NG_Avantgarde

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mtu NautIQ **BlueVision NG**

General Information

BVNG is MTU's ship automation system

Applications: Commercial and pleasure craft with Series 2000 / 4000 diesel engines

1-4 shafts and 1-6 control stands

■ Configurations

- MTU MCS* and MTU RCS** for FPP applications
- MTU MCS and interface to non MTU RCS for FPP, WJ and CPP applications
- Non MTU MCS and MTU RCS
- Non MTU MCS and non MTU RCS

Classification

- Non-classified: **mtu** NautIQ BlueVision NG_Basic
- Classified: **mtu** NautIQ BlueVision NG_Advanced
- Classified: **mtu** NautIQ BlueVision NG_Avantgarde



mtu NautIQ BlueVision NG

The modular system design of **mtu** NautIQ BlueVision allows for a flexible configuration: intelligent data technology ensures reliable data exchange and reduces the need for excessive cabling. Optimized interfaces between the propulsion and automation systems result in complete integrated solutions that guarantee security, efficiency and reliability – and all from one source.



Basic



Compact hardware

Compact size makes for easy installation and commissioning

Connected control

Of all components installed throughout the ship.

Local Operating Panel

For installation in the engine room.

Advanced



Type-approved components

LOP, control lever, display and instruments

Local Operating Panel

Comes with a color display and advanced functionalities such as clutch and speed control.

Data communication

Via redundant Ethernet ring bus.

Avantgarde



Type-approved components

LOP, control lever, monitor and instruments

A clear vision

The sophisticated color screen gives you a crystal-clear overview of what's happening on your propulsion system.



mtu NautIQ BlueVision NG_Avantgarde

mtu NautIQ BlueVision NG_Avantgarde provides the most sophisticated and extensively developed solution for standard automation systems. So you get a complete package where everything is just right: not only powerful engine performance, but also maximum efficiency, uncompromising reliability and environmental compatibility in monitoring and control of the complete propulsion plant.



A sophisticated solution

The most sophisticated and extensively developed solution for standard propulsion automation and includes a monitoring and remote control system package for your **mtu** engines and systems.

Type-approved components
LOP, control lever, monitor and instruments

Connected control
Of all components installed throughout the ship.

Compact hardware
Compact size makes for easy installation and commissioning.

A clear vision
The sophisticated color screen gives you a crystal-clear overview of what's happening on your propulsion system.

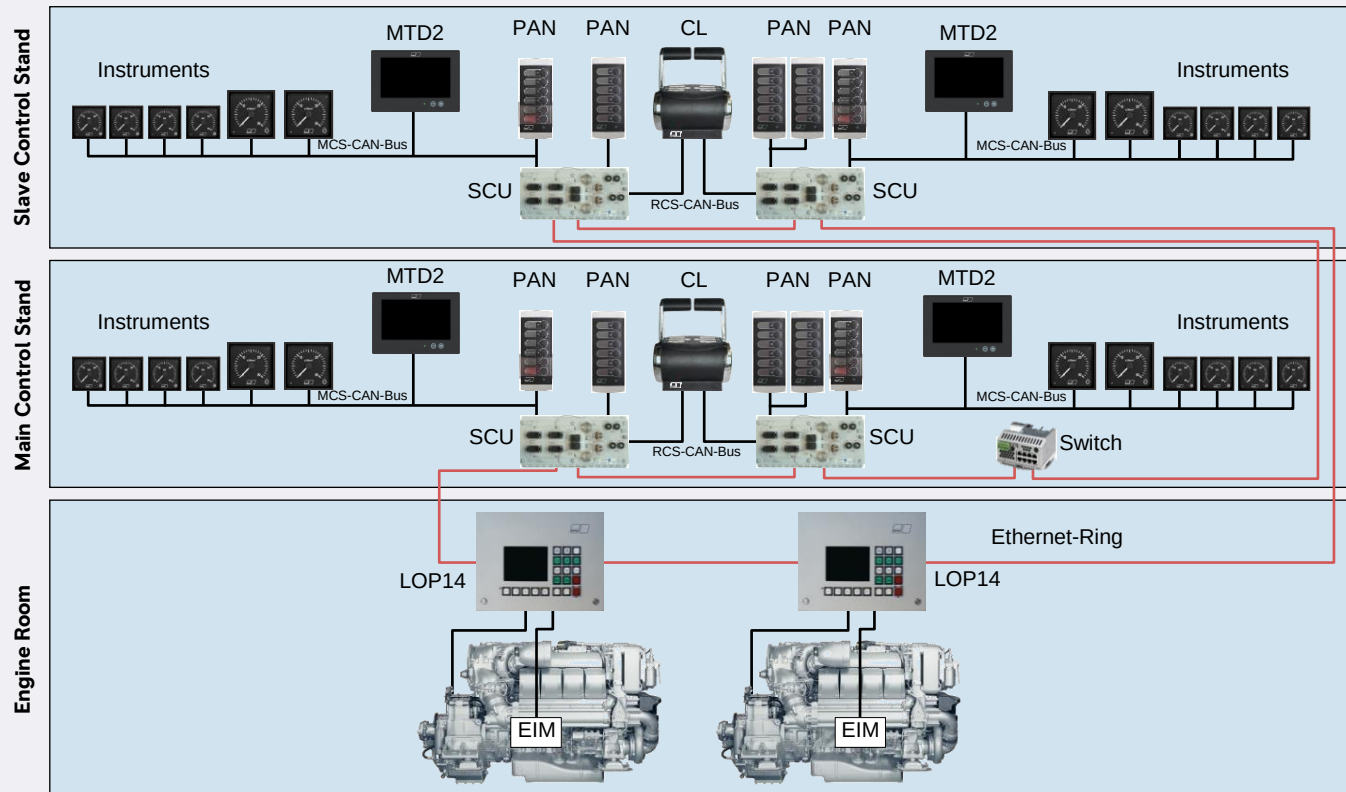


mtu NautIQ BlueVision NG_Advanced

System Layout –

Typical Configuration for BlueVision_Advanced

mtu NautIQ BlueVision
NG_Advanced is the basis for
mtu NautIQ BlueVision
NG_Avantgarde.



mtu NautIQ BlueVision NG_Avantgarde

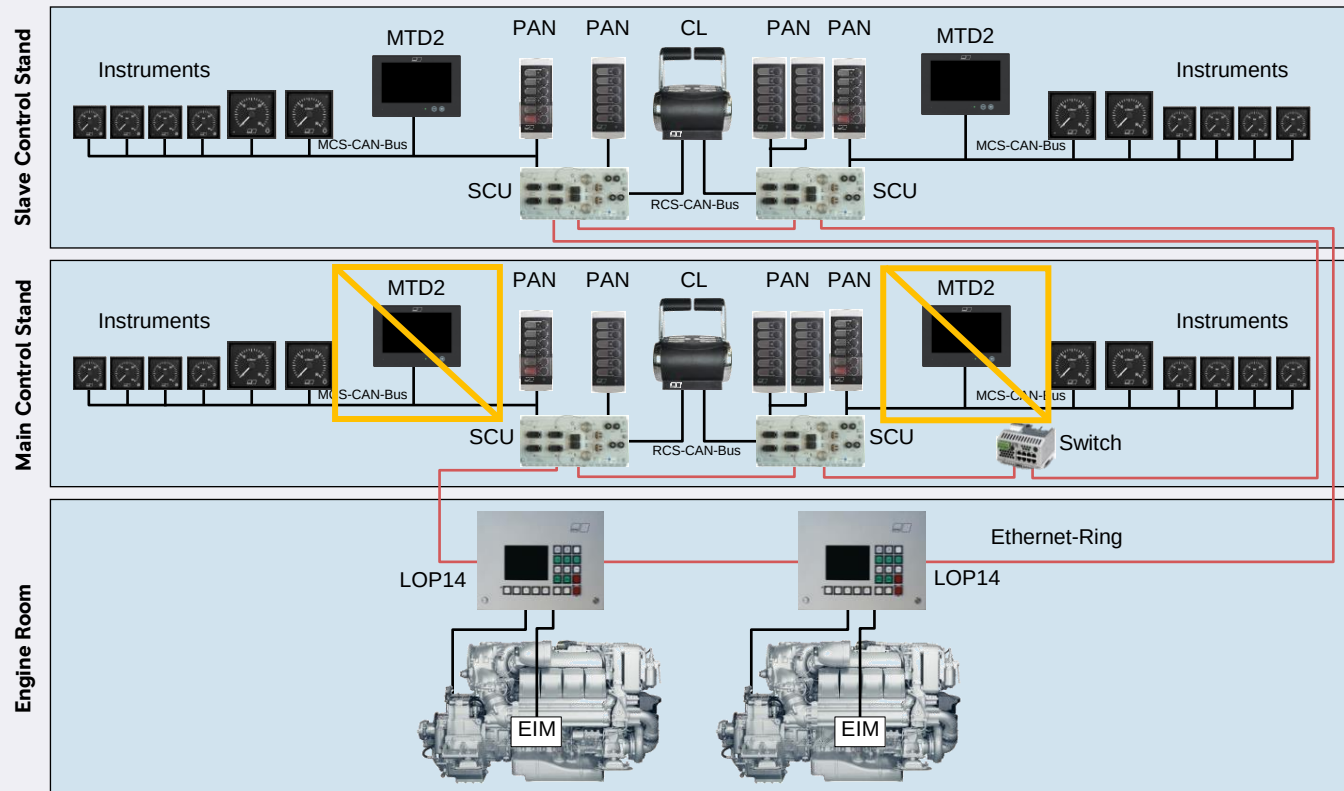
Update (1)

On Main Control Stand the Display (MTD2) will be replaced by operator station.

On Slave Control Stand 1 are 3 options available:

1. Operator Station
2. Display (MTD2)
3. Without Operator Station and Display (MTD2)

For all other slave control stands, only the selection with or without display (MTD2) is possible.





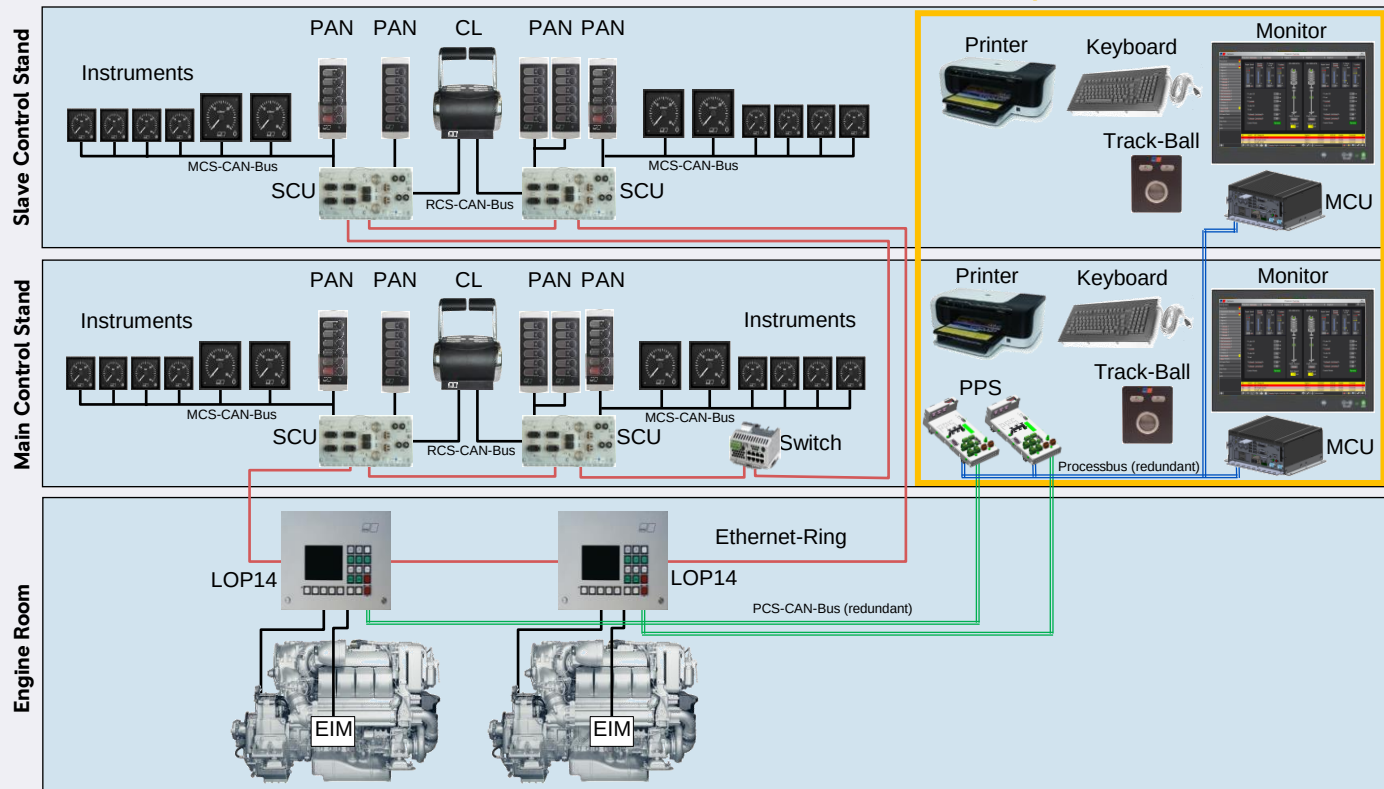
mtu NautIQ BlueVision NG_Avantgarde

Update (2)

The maximum possible scope for **mtu** NautIQ BlueVision NG_Avantgarde is for 4 shafts and 4 operator stations in total.

Maximum 2 operator stations per Control Stand.

Operator stations (see yellow frame) are just on Main and Slave Control Stand 1 available.





mtu NautIQ BlueVision NG_Avantgarde

Monitoring and Control Unit

Connections for:

- Monitor (Touch-Screen optional)
- Track-Ball
- Keyboard (optional)
- Printer for alarm and report printouts (optional)
- Programmable Process Station (PPS), Communication interface between field bus and process bus



Monitoring and Control Unit MCU 14

- Operator Station for the **mtu** NautIQ BlueVision NG_Avantgarde

Windows 10 operating system

- Visualization studio (Intouch 11.1)
- Introduction of new Monitor 16:9



mtu NautIQ BlueVision NG_Avantgarde

Technical Overview –
Monitor



Hatteland Display

Series X G2

- 16"; 16:9
- 24"; 16:9

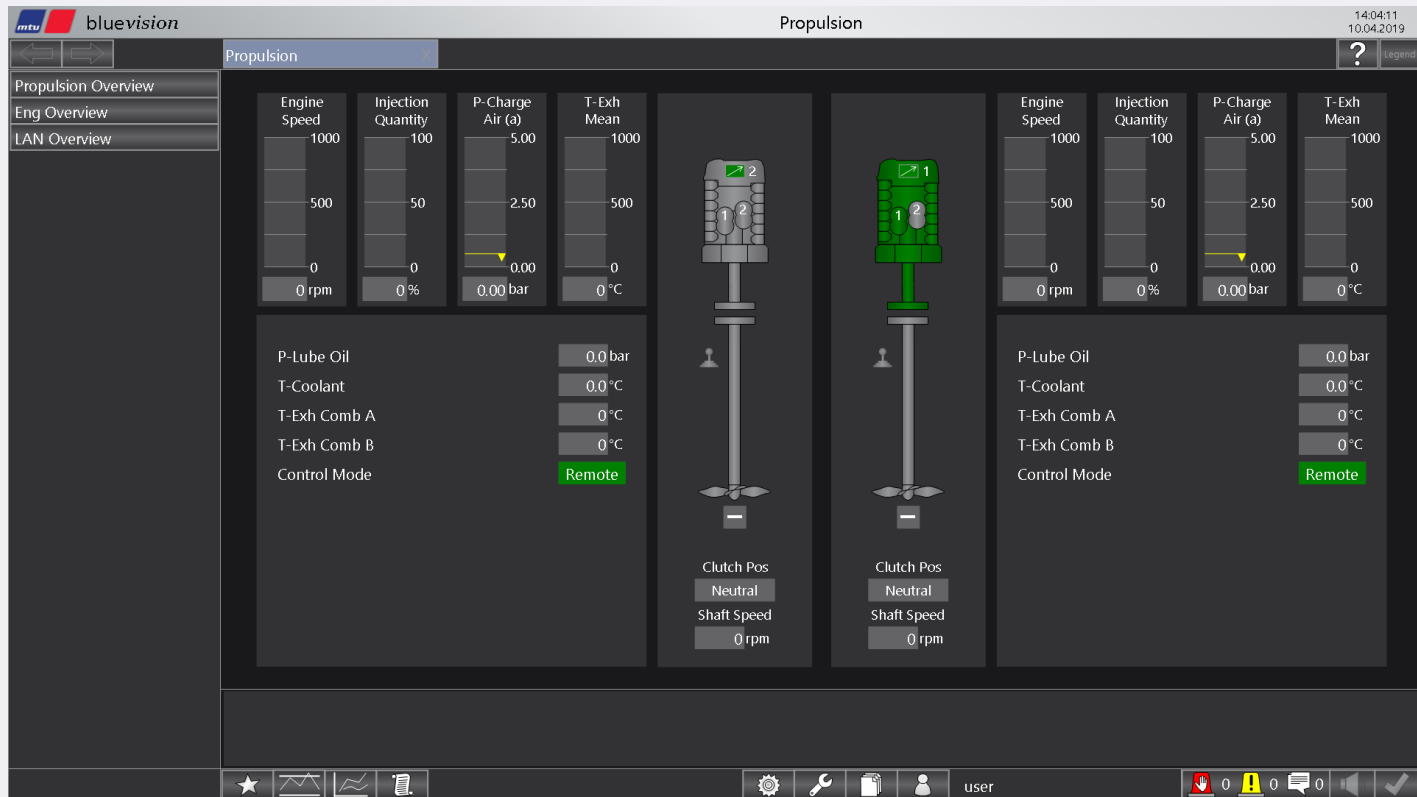
**All monitors are also
available as touch version.**



mtu NautIQ BlueVision NG_Avantgarde

Graphic Page –

Propulsion Overview





mtu NautIQ BlueVision NG_Avantgarde

Graphic Page –

Engine Overview





Cooling System

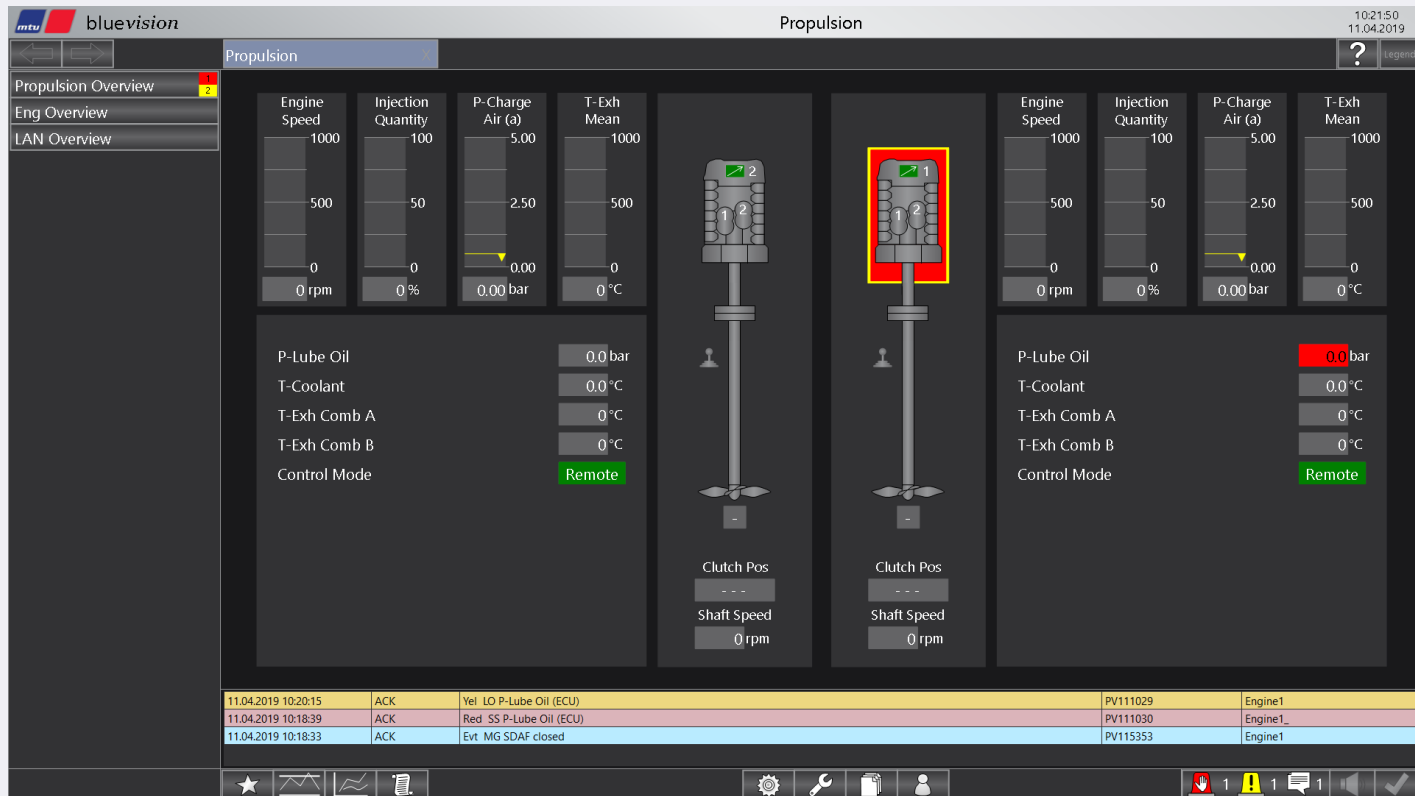




mtu NautIQ BlueVision NG_Avantgarde

Graphic Page –

Alarm indication on
Propulsion Overview





mtu NautIQ BlueVision NG_Avantgarde

Graphic Page -

Alarm page

mtu bluevision

Alarm Page

10:25:30
11.04.2019

Propulsion Overview 13
Eng Overview 37
LAN Overview

Date and Time

State

Measuring Point

Tagname

Alarm Group

11.04.2019 10:24:49	UNACK	Yel HI T-Exhaust A9	PV123309	Exhaust2
11.04.2019 10:24:49	UNACK	Yel HI T-Exhaust A8	PV123308	Exhaust2
11.04.2019 10:24:49	UNACK	Yel HI T-Exhaust A7	PV123307	Exhaust2
11.04.2019 10:24:49	UNACK	Yel HI T-Exhaust A6	PV123306	Exhaust2
11.04.2019 10:24:49	UNACK	Yel HI T-Exhaust A5	PV123305	Exhaust2
11.04.2019 10:24:49	UNACK	Yel HI T-Exhaust A4	PV123304	Exhaust2
11.04.2019 10:24:49	UNACK	Yel HI T-Exhaust A3	PV123303	Exhaust2
11.04.2019 10:24:49	UNACK	Yel HI T-Exhaust A2	PV123302	Exhaust2
11.04.2019 10:24:49	UNACK	Yel HI T-Exhaust A1	PV123301	Exhaust2
11.04.2019 10:24:49	UNACK	Yel HI T-Bearing Mean	PV123272	Bearing2
11.04.2019 10:24:49	UNACK	Yel AL EMU TranOut 2 Short Circuit	PV123191	Engine2_AI
11.04.2019 10:24:49	UNACK	Yel AL EMU TranOut 2 Open Load	PV123181	Engine2_AI
11.04.2019 10:24:46	UNACK	Red SS T-Coolant (EMU)	PV123130	Engine2
11.04.2019 10:24:46	UNACK	Yel HI T-Coolant (EMU)	PV123129	Engine2
11.04.2019 10:24:46	UNACK	Red SS P-Lube Oil (EMU)	PV123030	Engine2
11.04.2019 10:24:46	UNACK	Yel LO P-Lube Oil (EMU)	PV123029	Engine2
11.04.2019 10:24:46	UNACK	Red SS Overspeed (EMU)	PV123006	Engine2
11.04.2019 10:24:46	UNACK	Yel AL MCR Overload	PV121390	Engine2_AI
11.04.2019 10:24:46	UNACK	Red SS ETC 4 Overspeed	PV121371	Engine2
11.04.2019 10:24:46	UNACK	Yel HI ETC 4 Speed	PV121370	Engine2
11.04.2019 10:24:46	UNACK	Red SS ETC 3 Overspeed	PV121369	Engine2
11.04.2019 10:24:46	UNACK	Yel HI ETC 3 Speed	PV121368	Engine2
11.04.2019 10:24:46	UNACK	Red SS T-Exhaust Combined D	PV121344	Engine2
11.04.2019 10:24:46	UNACK	Yel HI T-Exhaust Combined D	PV121343	Engine2
11.04.2019 10:24:46	UNACK	Red SS T-Exhaust Combined C	PV121339	Engine2
11.04.2019 10:24:46	UNACK	Yel HI T-Exhaust Combined C	PV121338	Engine2
11.04.2019 10:24:46	UNACK	Yel AL ETC 4 Cut in Fail	PV121312	Engine2
11.04.2019 10:24:46	UNACK	Evt MG System Testmode	PV115900	Engine1_AI
11.04.2019 10:24:46	UNACK	Yel AL Status Transistor Outputs	PV115603	Engine1_AI
11.04.2019 10:24:46	UNACK	Yel AL Power Amplifier 2 Failure	PV115602	Engine1_AI
11.04.2019 10:24:46	UNACK	Yel AL Power Amplifier 1 Failure	PV115601	Engine1_AI
11.04.2019 10:24:46	UNACK	Red AL Shut Down Air Flaps Closed	PV115494	Engine1_AI
11.04.2019 10:24:46	UNACK	Red SS Overspeed (ECU)	PV115492	Engine1

Alarm Printout

Alarm Control

Alarm View

Alarm Prio

Actual

All

Alarm Group

Alarm State

Combined

All

Number of Alarms

50

Filter

Filter off

Star

Line

Area

Bar

Settings

Tools

Help

Users

13

36

1

Speaker

Check



***mtu* NautIQ systems will look
after your platform leaving
you to focus on the task**

Thank you for your attention!